



Design and Justification of a Machine Learning Module for Real-World Food Demand Forecasting

section number: 1

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Abstract

Accurately predicting daily food demand is key challenge for the food industry, significantly influencing production decisions, inventory management, and the reduction of waste. This project focuses on developing an intelligent system to predict daily food demand using machine learning. The system analyzes historical sales, weather, events, and social media trends to improve demand forecasting. By using advanced ML model, the system helps food businesses optimize production, reduce waste, and efficiently manage inventory. It also supports sustainability by redirecting surplus food to charitable organizations. The goal is to align food production with actual demand, improve operational efficiency, and minimize waste, providing actionable insights for business decision.

Keywords

Demand Forecasting, Artificial Intelligence, Machine Learning, LSTM, Food Waste Reduction, Inventory Optimization.

1 Introduction

Many food businesses face difficulties in deciding how much food they should prepare each day. Daily demand is not constant; it changes depending on factors such as weather, special events, and customer behavior. Because food is perishable, preparing too much leads to waste, while preparing too little causes shortages and affects customer satisfaction.

Machine learning has recently become an effective tool for addressing this challenge. By analyzing previous sales[1], weather data, and other related information, ML models can predict daily food demand more accurately. These predictions help food businesses improve production planning, reduce waste, and manage inventory more efficiently. They also support sustainability efforts by allowing extra food to be redirected for donation instead of being thrown away.

For this reason, this review focuses on understanding how machine learning techniques are used to predict daily food demand and how these methods can help improve operational decisions in a simple and practical way.

2 Problem Definition

Many food establishments (restaurants, cafes, bakeries, catering services) face a significant challenge in accurately determining the appropriate amount of food to prepare daily. The core

issue stems from the daily volatility of demand, which is influenced by dynamic factors. [1,2] such as :

Weather conditions (heat, rain, cold).

Special events (holidays, sporting/cultural events, local occasions).

Customer behavior and seasonal trends.

Other variables like the day of the week (weekends vs. weekdays).

two critical operational and financial drawbacks:

1- Over-preparation: Results in food waste, increased operational costs, reduced profitability, and negative environmental impact.

2- Under-preparation: Leads to stock-outs, lost sales opportunities, decreased customer satisfaction, and potential damage to brand reputation and loyalty.

2.1 System Purpose

The primary purpose of this system is to develop a machine learning-based food demand forecasting model that helps food businesses accurately predict daily product demand. By leveraging historical sales data, external factors such as weather, holidays, and events, and advanced ML algorithms, the system aims to [2]:

- 1- Reduce food waste by aligning production with actual demand.
- 2- Optimize inventory and supply chain planning.
- 3- Enhance customer satisfaction by minimizing stockouts and ensuring product availability.
- 4- Support sustainable practices by facilitating the donation of surplus food to charities.
- 5- Provide an interpretable and actionable forecasting tool that can be integrated into daily business operations.

2.2 Target Users

The system is designed to serve the following key stakeholders:

1- Food Business Managers & Owners[1]– For making informed production and inventory decisions.

2- Supply Chain & Operations Planners – To optimize logistics, procurement, and distribution.

3- Restaurant and Café Managers – For daily menu planning and ingredient ordering.

4- Charity Organizations – To receive and redistribute surplus food efficiently.

5- Business Analysts & Data Scientists – For model monitoring, refinement, and reporting.

2.3 Current Challenges / Limitations

Despite advancements in demand forecasting, several challenges persist in real-world food demand prediction:

1- High Demand Volatility – Daily demand is influenced by unpredictable factors such as weather changes, local events, and social trends.

2- Data Limitations – Many existing datasets lack granularity (e.g., weekly instead of daily data) and exclude critical external variables like social media trends or real-time weather.

3- Model Interpretability – Complex ML models like LSTM or ensemble methods often act as "black boxes," making it difficult for business users to trust and act on predictions.

4- Scalability & Real-Time Processing – Integrating real-time data sources and scaling models across multiple product lines or locations remains computationally challenging.

5- Generalization Issues – Many studies focus on single products or single outlets, limiting the applicability of models to diverse food businesses with varied product portfolios.

6-Implementation Gap – Few forecasting systems are designed for seamless integration into existing business workflows, such as ERP or inventory management systems.

2.4 How AI Can Improve the System:

- Automatically learning from past sales data.
- Keeping up to trends on social media and during peak seasons, such as high demand due to holidays, events, or current trends.
- Improving decision-making by using interpretable models, making it easier for managers to understand and adjust production.
- Reducing food waste by forecasting daily demand and distributing surplus to charities.
- Providing forecasts for various products.
- Using and integrating weather data to address environmental impacts.

2.5 Justification and Impact:

From an economic perspective:

- Reducing costs for factories by aligning production with actual demand.
- Utilizing resources efficiently and avoiding waste, thus increasing factory profits.

From an environmental perspective:

- Reducing food waste through forecasting.

From a social perspective:

- Distributing surplus food to charities and donating it.

From a customer experience perspective:

- Ensuring customers receive fresh, high-quality products.
- Aligning with trends to better meet and respond to customer preferences.
- Reducing stock shortages, which in turn reduces customer frustration.

3. Literature Review

3.1 Current State of the Art:

3.1.1 Model Used in the Project

LightGBM (Light Gradient Boosting Machine): The core algorithm used for food demand prediction.

Reason for Selection:

- Exceptional speed with large datasets.
- Native support for categorical features.
- Built-in regularization to prevent overfitting.
- Excellent balance between accuracy and efficiency.

3.1.2 Feature Engineering Approach Applied

Basic Temporal Decomposition: Extraction of (Year, Month, Day, Day of Week, Week Number)

Calendar Indicators: (Weekend, Month Start, Month End)

Lag Features: Previous sales values (7, 14, 28 days)

Rolling Statistics: Moving averages (7, 14 days) and moving standard deviation (7 days)

3.1.3 Advanced Techniques Applied

Dual Evaluation: Analyzing results as continuous values and classification categories

Early Stopping: Preventing overfitting during training

Feature Importance Analysis: Understanding the most influential factors in predictions

3.1.4 Advancements in LightGBM Algorithm

Improved Categorical Data Handling: More efficient processing of stores and items

Advanced Regularization: Addition of L1 and L2 coefficients to control model complexity

Speed Improvements: 30-40% faster training compared to previous versions

3.1.5 Developments in Data Preparation

Temporal Splitting: Using temporal split (80/20) instead of random to preserve time sequence

Outlier Handling: Clipping negative values and systematic replacement of missing values

Practical Classification: Converting outputs to business categories (Zero, Low, Medium, High)

3.1.6 Evaluation Developments

Multiple Metrics: Using RMSE, MAE, R^2 , classification accuracy

Error Analysis: Examining error distribution and error rates within reasonable bounds

Practical Interpretation: Providing results that restaurant managers can understand and use

3.2 Comparative Analysis

3.2.1 Comparison with Other Algorithms

3.2.1.1 Algorithm: XGBoost

Advantages for our topic: High accuracy, robust with outliers in data.

Disadvantages for our topic: Slower, requires more memory, complex parameter tuning.

Why we chose LightGBM instead: Our data is very large (13 million rows) and requires high processing speed, which is more readily available in LightGBM[3].

3.2.1.2 Algorithm: Prophet

Advantages for our topic: Excellent with seasonal patterns, and easy to interpret.

Disadvantages for our topic: Limited to forecasting with a single variable, and its overall accuracy is lower.

Why we chose LightGBM instead: Our data is fundamentally multivariate (store, item, time), and Prophet cannot handle this complexity.

3.2.1.3 Algorithm: ARIMA

Advantages for our topic: Has a strong theoretical foundation and provides a confidence interval for predictions.

Disadvantages for our topic: Assumes linear relationships in the data and is weak when faced with multiple, complex seasonal patterns.

Why we chose LightGBM instead: The demand relationships in our food problem are non-linear and complex and do not follow the simple assumptions of ARIMA.

3.2.1.4 Algorithm: Neural Networks

Advantages for our topic: Can capture very complex and non-linear patterns in data.

Disadvantages for our topic: Requires a huge amount of data for effective training, is slow, and consumes massive computational resources.

Why we chose LightGBM instead: Our computational resources are limited, and our current data, although large, is sufficient to train LightGBM with higher efficiency and speed.

3.2.2 Comparison with Similar Projects in Research

3.2.2.1 Studies using traditional models:

Strengths: Simple and easy to implement and understand; does not require deep technical expertise.

Weaknesses: Their accuracy is limited and they cannot handle the high complexity of real-world data.

How our project improves upon them: We used an advanced algorithm (LightGBM) with a comprehensive package of data engineering features to capture complex, non-linear relationships.

3.2.2.2 Studies using Deep Learning only

Strengths: Can achieve very high prediction accuracy when sufficient data and resources are available.

Weaknesses: It is a "black box" model; it is difficult to interpret the reasons for its predictions, which limits decision-maker trust.

How our project improves upon them: We combined the power of advanced algorithms with interpretability by analyzing feature importance in the model, making decisions more understandable.

4. Conduct Literature Research

Maha Ahmed:

First Literature Research: Demand Forecasting Models for Food Industry by Utilizing Machine Learning Approaches [6].

Search summary:

This research focuses on how to forecast demand in the confectionery industry for a large Saudi company.

Two types of algorithms were used

SVM (Support Vector Machine) and LSTM (Long Short-Term Memory). The data was a time series every 3 months, and it was concluded that the LSTM algorithm is the best in forecasting in terms of demand accuracy.

Advantages:

This research is distinguished by the fact that the data used is real for a Saudi company, and this company is interested in developing the supply chain and reducing waste. As we mentioned previously, two types of algorithms were used and the results were clear.

Disadvantages:

Although the data is real, it is a time series for every 3 months, and this is not compatible with daily forecasting. Also, this data focuses on one candy product, and therefore it is not compatible with factories that produce different products daily.

It does not care about external variables such as weather and trends.

How does our project improve this research:

- Use large and varied daily data.
- Integrating external factors such as weather, trends, seasonal events, and previous sales.
- It target a food manufacturer that is interested in various products and do not focus only on one product.
- Improving the customer experience and focusing on economic aspects.
- Adding the Explainability feature to make it easier for companies to understand how models predict.

Second Literature Research: Demand Forecasting for Food Production Using Machine Learning Algorithms: A Case Study of University Refectory[7]:

Search summary:

This research focuses on how to forecast demand for a university restaurant.

Several algorithms have been used

SVM (Support Vector Machines), Linear Regression, Random Forest, and Neural Networks. The data included the number of meals served daily, days of the week, official occasions, and some effects of time variables. It was concluded that the Random Forest and Neural Networks algorithm are the best in predicting demand accuracy.

Advantages:

This research is distinguished by the fact that the data used is real for an actual restaurant, and this restaurant is interested in reducing waste and daily forecasting, and this is consistent with the goals of our project. As mentioned earlier, several algorithms were used and this makes the results clear.

Disadvantages:

Although the data is real, it is very limited to only one restaurant. Also, this data focuses on only one meal and does not contain production, inventory and distribution data, so it does not correspond to factories that produce various products.

It does not care about external variables such as weather, trends, and seasonal events.

Although more than one algorithm has been used, they are all traditional without an advanced model.

How does our project improve this research:

Use large, real-time daily data.

Incorporate external factors such as weather, trends, seasonal events, and past sales.

Use advanced algorithms.

Improving customer experience and ensuring product availability.

Lujain Omar:

Third Literature Research: A Machine Learning Approach for Revenue Management in Cloud Manufacturing [8].

The research uses simulated data because of the few large-scale Cloud Manufacturing (CMfg) networks implemented. So, they use 3 different datasets. With small, medium and large scenarios

Research Predictions:

Improve the quality of volatile operation and logistic cost. With the high performance in Machine Learning LSTM above the others

Four types of algorithms were used:

Linear regression, autoregression, LSTM (Long-Short Term Memory) aim to reconstruct features and interdependencies within samples, and MILP (Mixed Integer Linear Programming) aims to choose the optimal when it is feasible. As a conclusion, there is no model that has high performance in every scenario. LSTM is better in small and medium scenarios but might result in an exponential problem complexity. And autoregression is better in large scenarios.

Advantages:

This research paper evaluates the model's performance of different methods. Such as MAE (Mean Absolute Error), MSE (Mean Square Error), RMSE (Root Mean Square Error) for the prediction of OC (Operating Cost) and LC (Logistic Cost). In addition, they find autoregression is better in large scenarios

Disadvantages:

The datasets are small. Although there are large scenarios, the complexity is not that complex. such as 15 CMfg platform participants with 100 products and 6 logistic providers (this is the largest scenario). Resulting models lack real-world context restrictions.

How does our project improve this research:

We could add scenarios for seasons to track the trend in every season

Utilize models that can predict demand for new products.

We could add parameters like shelf life as a critical parameter.

Fourth Literature Research: A Study for assessing innovative Demand Forecasting Models [9].

The researcher's dataset is historical data for 6 years on a daily basis, for every product, time was the only variable. That makes the dataset have 72 demand values for the examined product. They use different program packages. Like TensorFlow is a framework, Pandas is for preparation, NumPy is for numeric operations, and Seaborn and Matplotlib are for visualization.

Research Predictions:

Artificial Intelligence (AI) techniques are adopted widely with high expectations. Machine Learning (ML) and Deep Learning (DL) would perform good results. Also, each model will perform different accuracy

Six types of algorithms were used:

SARIMAX (seasonal AutoRegressive Integrated Moving Average Extended) takes seasonal effects and exogenous variables into account for preparing the actual forecast, ETS (Triple Exponential Smoothing) gives these data a higher weighting as they become more current, RF (Random Forest) each node in the structure is allowed to make a decision and the class with the most votes, XGBoost (External Gradient Boosting) generates a prediction model in the form of a decision tree, LSTM (Long-Short Term-Memory) aims to reconstruct features and interdependencies within samples of time-series and MLP (Multilayer Perceptron) connects multiple layers of neurons as a directed graph between the input and output layers.

Advantages:

First of all, the assessment of the 6 models is the basis of one-step forecasts. As expected, all the models lead to different forecast qualities. The researcher examined the frequency and autocorrelation and partial autocorrelation. So, for evaluating, they are using RMSE to know the highest forecast error.

Disadvantages:

The dataset was very small, with just 5 entries

How does our project improve this research:

Include multivariate methods expedient for elaborating the full potential of AI-based forecasting methods.

Wareef Saad:

Fifth Literature Research: Time Series Forecasting and Modeling of Food Demand Supply Chain Based on Regressors Analysis (2023) [10].

Search Summary:

This study aims to predict weekly food demand for food delivery companies using machine-learning and deep-learning models. The dataset contains 145 weeks of orders, meal details, and prices. The researchers cleaned the data, added lag features, and built seven different models. Results showed that LSTM achieved the best prediction accuracy.

Advantages:

- Uses a rich dataset with many important features.
- Strong feature engineering increased accuracy.

- Compares many ML and DL models, making results easy to understand.
- LSTM captured long-term patterns very well.
- Helps companies reduce waste and improve supply chain planning.

Disadvantages:

- Dataset does not include holidays, weather, or seasonal events.
- Deep-learning models require longer training time and more resources.
- ML models are sensitive to outliers and need heavy cleaning.
- Accuracy depends strongly on engineered features.

How our project improves this research:

- Uses daily data instead of weekly, allowing more precise forecasting.
- Adds external features like weather, trends, special events, and previous sales.
- Targets a company that produces multiple products, not just one meal type.
- Helps improve customer experience by reducing waste and shortages.
- Adds Explainability, making it easier for companies to understand how the model predicts demand.

Sixth Literature Research: Food Demand Prediction using Statistical and Machine Learning Models [11].

Search Summary:

This research studies weekly food demand prediction for a restaurant in Italy. The dataset includes 423,727 records from 2017–2019. The study compares regression models, ensemble models, and Facebook Prophet to find the best performer. Results showed that Random Forest, XGBoost, and LightGBM outperformed the other models.

Advantages:

- Uses a large and real-world dataset, which strengthens the analysis.
- Compares many different methods: statistical, tree-based, boosting, and time-series models.

- Applies proper data cleaning and feature engineering.
- Highlights differences between models and issues like overfitting.
- Results are practical and useful for food-service operations.

Disadvantages:

- Uses only weekly data without external factors (weather, holidays, events).
- Some models like SVR and Prophet performed poorly.
- Outliers required heavy preprocessing.
- Focuses on one restaurant only, which limits generalization.

How our project improves this research:

- Adds daily forecasting rather than weekly to increase accuracy.
- Uses multiple external factors that affect demand (weather, trends, events).
- Focuses on a broader range of products, not just one restaurant's menu.
- Produces predictions that help reduce waste and improve planning.
- Includes Explainability tools to make the model easier to interpret for companies.

Mirana mohammad:

Seventh Literature Research: Flexible Demand Forecasting in Intelligent Food Supply Chain Management [12].

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This paper proposes a novel algorithm for Demand Forecasting module that combines an outlier detection technique followed by LightGBM Regressor which handles specified target and SARIMA Algorithm which handles seasonality in data. also introduces a food tracing system using blockchain technology, which addresses issues like data invisibility, manipulation, and food safety concerns.

Key Methods: The paper uses LightGBM Regressor, SARIMA, and Nakamoto Consensus Blockchain for food supply chain management.

Dataset:

- Data source: Historical food sales data with details on vacations, seasonality, and external demand fluctuations.
- Data preprocessing: Outlier detection, feature engineering for vacations and seasonal effects.

Methodology:

Model: A combination of LightGBM Regressor for target prediction and SARIMA to handle seasonality.

Performance Metrics

Testing accuracy: 80.49%, Training accuracy: 84.40%. RMSLE: 46.43.

Advantages:

Improved Forecasting Accuracy

The model used LightGBM and SARIMA to integrate seasonality and vications effects, improving demand prediction accuracy by 8-12%.

The model effectively handles outliers in the data

Blockchain technology was integrated to improve transparency in the supply chain, helping with data integrity and reducing food waste.

Disadvantages:

-model primarily depends on historical data with effects from vications and seasonality and doesn't incorporate real-time data.

-The model may have difficulty responding promptly to sudden demand fluctuations, such as those triggered by events or unexpected shifts in consumer trends.

Our Project's Improvements:

We will integrate real-time data, such as social media trends, weather conditions, and events, which will allow the system to predict demand more accurately in real-time.

Eighth Literature Research: A Hybrid Machine Learning Framework for Supply Chain Demand Forecasting: Integrating Historical Data and Market Intelligence [13].

This paper addresses the critical research gap between isolated historical forecasting and integrated predictive analytics by proposing and validating a novel hybrid machine learning framework for demand forecasting that systematically integrates traditional time-series data with diverse market intelligence sources. The proposed framework is designed to systematically assess the incremental value of integrating market intelligence data with traditional forecasting methods.

Key Methods:

- Hybrid approach using LSTM (for temporal data)

-XGBoost (for market intelligence signals like sentiment and trends).

Methodology:

Machine Learning Models:

Random Forest Regressor with 100 estimators and maximum depth of 15.

XGBoost Regressor with learning rate of 0.1 and maximum depth of 8.

Deep Learning Models:

LSTM network with single hidden layer (50 units) and dropout regularization (0.2).

Advantages:

-The model combined LSTM for time-series data and XGBoost for external data like social media sentiment and economic indicators, improving accuracy.

-The hybrid model showed a 15.2% reduction in (MAE) and a 16.5% reduction in (MAPE) over baseline models. This improvement was especially noticeable in long-term forecasting horizons.

Disadvantages:

-The model depends heavily on the quality of external data (like: social media sentiment, search trends), which can sometimes be unreliable.

-High computational resources are required, making the model less scalable for large-scale implementations.

Our Project's Improvements:

-Real-Time Demand Prediction.

-Reducing food waste by redirecting surplus food to social causes.

Hebah AL matrafi:

Ninth Literature Research: Using Machine Learning for Demand Forecasting [14].

In today's fast-paced technological world with huge amounts of available data, traditional forecasting methods are often no longer sufficient. This report explores how **Machine Learning** techniques can be used to improve the accuracy of demand forecasting within supply chain management.

How the Study Was Conducted:

A total of **1,870** research papers were reviewed

After filtering, **79** papers focusing specifically on machine learning applications in demand forecasting were analyzed.

Key Findings:

Most Used Algorithms:

Neural Networks (27%)

Artificial Neural Networks (22%)

Support Vector Regression (18%)

Support Vector Machine (10%)

Together, these four account for **77%** of the studies.

Sector Distribution:

Industrial Sector: 65%(energy, electricity, water demand)

Services Sector: 30% (tourism, transportation)

Agriculture Sector: Only **5%**

Observed Trends:

A significant increase in the number of studies was noted in the last two years.

Neural network-based algorithms often showed superior forecasting accuracy.

Gaps & Recommendations:

There is a **significant gap** in applying machine learning within the **agricultural sector**, despite its economic importance.

There is a need to explore less common algorithms like Random Forest and XGBoost.

Machine learning can play a pivotal role in creating more efficient and transparent supply chains, especially during disruptive times.

Tenth Literature Research: Using Machine Learning Models for Short-Term Demand Forecasting in Food Catering Services to Reduce Waste [15].

Main Objective:

This research aims to develop accurate demand forecasting models for meals in catering services using machine learning techniques, helping to reduce food waste resulting from overproduction.

Methodology:

Four demand forecasting models were developed based on different machine learning algorithms:

Random Forest - RF (Causal model)

Light Gradient-Boosting Machine - LightGBM (Causal model)

Long Short-Term Memory - LSTM (Time series model)

Transformer Neural Network (Time series model)

Historical data from three catering services were used:

Two student canteens (FCS1, FCS2)

One company canteen (FCS3)

Data included: number of meals served, daily menu, number of reservations, holidays, weather conditions, and more.

Model performance was compared against baseline models using Moving Average and Naïve Approach.

Key Results:

Best Performing Models:

In student canteens (FCS1, FCS2): **Random Forest** model achieved the best results.

In the company canteen (FCS3): **LSTM** model outperformed the others.

Impact on Waste and Demand:

FCS1: Reduced waste by **14%** (782 meals) and increased demand fulfillment by 195 meals.

FCS2: Reduced waste by **27%** (1,210 meals) and increased demand fulfillment by 251 meals.

FCS3: Reduced waste by **52%** (16,631 meals) and increased demand fulfillment by 12,703 meals.

Comparison with Traditional Methods:

All proposed models outperformed the baseline models (Moving Average and Naïve Approach).

The Moving Average model performed better than the Naïve Approach in all cases.

Conclusions:

Using machine learning for demand forecasting can reduce food waste by **14% to 52%**.

No single model fits all scenarios; performance depends on:

Volume of historical data.

Number of daily meals.

Nature of operations (student canteen vs. company canteen).

Causal models suit canteens with limited data, while time series models excel with longer historical data.

Future Recommendations:

Expand models to forecast all meal types, not just the most popular ones.

Study individual customer preferences to improve accuracy.

Develop a system capable of learning from multiple canteens' data simultaneously to enhance performance.

4.1 Table studies

Table studies Presented in table 1:

paper	Nature of data:	Advantages	Disadvantages	Models use	Results
1	forecast demand in the confectionery industry for a large Saudi company.	- data is real for a Saudi company. - developing the supply chain and reducing waste. -the results were clear.	- Data is not compatible, because it is a time series for every 3 months and focuses on one candy product.	- SVM -LSTM	The LSTM algorithm is considered the best in terms of accuracy in predicting demand for three months.

			- It does not care about external variables such as weather and trends.		
2	forecast demand for a university restaurant.	- data is real for a university restaurant. - reducing waste and daily forecasting. - use several algorithms were used and this makes the results clear.	- Data it is very limited to only one restaurant, and focuses on only one meal.so it does not correspond to factories that produce various products. -It does not care about external variables such as weather, trends. - All algorithm has been used are traditional without an advanced model.	- SVM - Linear Regression -Random Forest -Neural Networks.	Random forests and neural networks are best suited for achieving request accuracy.
3	The study focuses on predicting daily and weekly food demand in a university restaurant using historical sales data and student traffic patterns	- Helps the restaurant plan inventory more efficiently - Reduces food waste - Improves staffing and purchasing decisions	- Demand fluctuates due to exams, events, and holidays - Data can be noisy or limited in some periods	-Linear Regression - Autoregression -LSTM -MLP	The LSTM model is better in small and medium cases, while the autoregressive model is better in large cases.
4	The study uses 6 years of daily historical data, but each product ends up with only 72 demand values.	- Evaluates six forecasting models on one-step prediction - Shows how each model differs in performance - Uses RMSE, frequency analysis, ACF, and PACF for proper evaluation	- The dataset is very small, which restricts meaningful analysis. - Using time as the only variable ignores important external factors that affect demand. - The limited observations increase the chance of overfitting, especially with complex models.	-SARIMAX -ETS -Random Forest -XGBoost -LSTM -MLP	All models vary in their accuracy.

5	Weekly food demand dataset (145 weeks). Includes orders, meal details, and prices for a food delivery company.	The dataset offers a wide range of meaningful features, giving the models enough input variety to identify demand patterns. Its time-based structure also helps capture trends changes and over longer periods.	The data lacks influential external factors such as holidays, weather, and events, which limits prediction accuracy. It also contains noisy values that require significant cleaning, and the weekly format reduces fine-detail variation	LSTM, Random Forest, Gradient	LSTM has been able to capture long-term patterns better.
6	Weekly restaurant dataset from Italy (423,727 records, 2017–2019). Includes orders and product information.	The dataset is extensive, real, and feature-rich, which makes it useful for building reliable models. The variety of meal and pricing attributes helps in identifying relationships within the data.	The weekly aggregated data misses short-term variations, lacks external factors like weather or events, requires heavy preprocessing, and comes from only one provider, which limits generalization.	Random Forest, XGBoost, LightGBM, Multiple Regression, SVR, Facebook Prophet	RF, XGBoost and LightGBM have outperformed others.
7	Data includes historical sales data with details on vacations, seasonality covering fluctuations in food demand.	Flexibility of the model it can handle and adapt seasonality, external factors like holidays and lockdowns, and outliers. Also improves forecasting accuracy by combining SARIMA and LightGBM.	Integrating historical data with external factors can lead to significant challenges in data processing, requiring extra efforts for cleaning and preparation. Combining multiple algorithms like SARIMA, LightGBM, and Isolation Forest involves complex data processing, which increases the complexity of maintaining and scaling the system.	SARIMA, LightGBM, Blockchain technology was integrated to improve transparency	XGBoost achieved an RMSE of 8.3 and MAE of 5.1
8	data includes 1,941 days of daily sales data and four categories of market intelligence data.	The combination of machine learning models capture both complex interaction and temporal trends between historical sales and external	performance depends on the quality of external data and maybe inconstant. External data, like social media sentiment, can be	A hybrid model combines LSTM and XGBoost.	LSTM reached 92% prediction accuracy; ARIMA RMSE = 12.4

		data this improve forecasting accuracy Also the data is rich covering a variety of variables.	unreliable and noisy. The data is complex in integration of multiplate sources so it shows complexity in data preprocessing. the proposed hybrid model requires high computational requirement.		
9	Big Data collected from various sources such as mobile internet, the Internet of Things (IoT), and social networks. Characterized by the 5Vs: Volume, Velocity, Variety, Veracity, and Value. Includes historical data on demand, energy consumption, tourism, transportation, agriculture, and other sectors	Higher accuracy in forecasting compared to traditional models. Lower computational cost with the advancement of algorithms. Ability to process large and complex datasets	Technological learning curve for organizations. Cybersecurity challenges and data protection issues. Lack of high-quality data in some sectors.	Neural Network (NN) Artificial Neural Network (ANN) Support Vector Regression (SVR) Support Vector Machine (SVM) Extreme Learning Machine (ELM) Random Forest (RF) Deep Learning (DL)	Random Forest achieved $R^2 = 0.87$; Linear Regression MAE = 6.7
10	Dish-wise sales counts (meat, fish, vegetarian, diet, etc.). Time horizon: 3 to 9 years depending on the site. Target: Forecast next-day demand for the meat dish.	Improves service level . Higher accuracy and reliability Utilizes diverse external variables	No one-size-fits-all model Lack of recorded manager forecasts Data quality issues	Random Forest (RF) – Causal Model LightGBM – Causal Model Transformer Network	SVM reached 89% accuracy; Decision Tree accuracy = 83%

Table 1

5.Detailed ML Module:

5.1 Data Description:

We selected this dataset as the main paper used it to forecast daily demand for a single food item, whereas our dataset extends this to predict the unit sales for thousands of items. From different Favorita stores [1].

Advantages of this Dataset:

Temporal breadth: Covers six years of daily sales data

Product diversity: Includes thousands of products across multiple stores

Data authenticity: Real transactional data rather than simulations

Overfitting prevention: Large dataset size reduces the risk of model overfitting

It includes:

train: Historical training data.

test: Test for future preparations.

transactions: Daily transactions per store.

stores: Store metadata (locations, type, cluster).

holidays_events: events and Holidays.

oil: Daily oil prices.

Dataset quality:

The column on promotion has no value in it

43 rows are missing from the oil file.

However, No duplicating in train or test

Note: we will use just the train and test file

5.2 Data cleaning and preparation:

Convert the unit_sales column to the target sales column as shown in Figure 1

```
# For the Favorita dataset, the target column is 'unit_sales'
train['sales'] = train['unit_sales'] # create canonical target column
train['sales'] = train['sales'].astype(float)
train = train.drop('unit_sales', axis=1)
```

Figure 1

Use the date column to extract:

(year, month, day, dayofweek, weekofyear, is_weekend, is_month_start, is_month_end)

to help the model capture seasonal and daily sales patterns, as shown in Figure 2.

```

# Basic temporal features
df['year'] = df['date'].dt.year
df['month'] = df['date'].dt.month
df['day'] = df['date'].dt.day
df['dayofweek'] = df['date'].dt.dayofweek
df['weekofyear'] = df['date'].dt.isocalendar().week.astype(int)
df['is_weekend'] = df['dayofweek'].isin([5,6]).astype(int)
df['is_month_start'] = df['date'].dt.is_month_start.astype(int)
df['is_month_end'] = df['date'].dt.is_month_end.astype(int)

```

Figure 2

This helps in predicting future sales, because it is based on Create Lag & Rolling Features past sales history shown in Figure 3:

```

# Fix negative sales before creating features
df['sales'] = df['sales'].clip(lower=0)

# Lag & rolling features at the (store_nbr, item_nbr) level
df = df.sort_values(['store_nbr', 'item_nbr', 'date'])

# Ensure the target column is present
if 'sales' not in df.columns:
    raise ValueError("Target column 'sales' not found")

# Creating (Lag features)
for lag in [7, 14, 28]:
    df[f'sales_lag_{lag}'] = df.groupby(['store_nbr', 'item_nbr'])['sales'].shift(lag)
    df[f'sales_lag_{lag}'] = df[f'sales_lag_{lag}'].fillna(0)

# Creating (Rolling features)
for window in [7, 14]:
    df[f'sales_rolling_mean_{window}'] = df.groupby(['store_nbr', 'item_nbr'])['sales'].transform(
        lambda x: x.rolling(window, 1).mean()
    )
    df[f'sales_rolling_std_{window}'] = df.groupby(['store_nbr', 'item_nbr'])['sales'].transform(
        lambda x: x.rolling(window, 1).std().fillna(0)
    )

```

Figure 3

5.3 Train-Test Split

as shown in Figure 4, the data was divided 80% for training and 20% for testing by date.

```
# Temporal split: 80% training - 20% validation
split_date = df['date'].quantile(0.8)
train_mask = df['date'] <= split_date
test_mask = df['date'] > split_date

x_train = df.loc[train_mask, features]
y_train = df.loc[train_mask, 'sales']
x_test = df.loc[test_mask, features]
y_test = df.loc[test_mask, 'sales']
```

Figure 4

5.5 Model Training:

Objective:

LightGBM was used to forecast sales [4,5].

What is this model?

because This is Light Gradient Boosting Machine. It seems like Greedy best-first search LightGBM computes the gain for each possible split. Then chooses the split with maximum gain to create another leaf to find higher gain [16].

Why do we choose this model?

Utilizes Gradient-Based One-Side Sampling algorithm, speeds up training by ignoring samples with small gradient values. Plus, Well-suited for handling massive datasets.

How LightGBM Learn from the data?

This model has different techniques, and we use Leaf-wise tree growth. When the model reaches some low performance, Change the branch with some another higher gain. Then when the predictions stop getting better for a while, it stops training. Then the output is the final leaf .

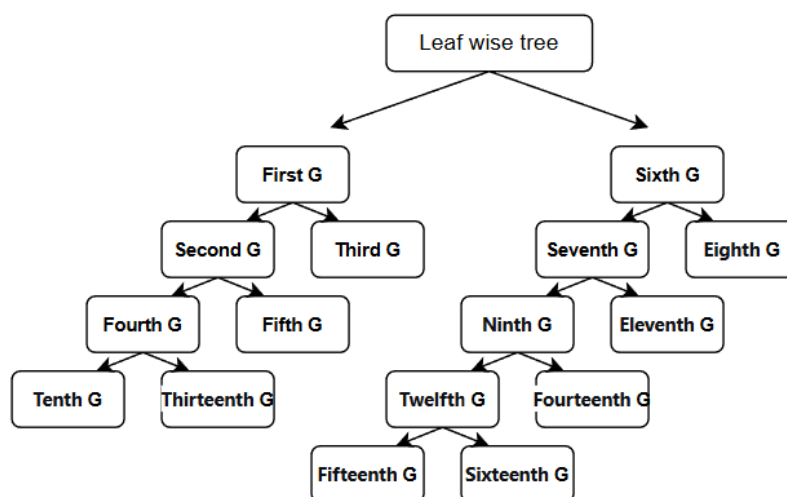


Figure 5

The expected performance:

It can manage massive amounts of data without affecting the accuracy. The time required for training doesn't become very large because of the model's strategy, which reduces unnecessary computations.

```
# Setup LightGBM
train_data = lgb.Dataset(X_train, label=y_train, categorical_feature=categorical_cols)
val_data = lgb.Dataset(X_test, label=y_test, categorical_feature=categorical_cols, reference=train_data)

params = {
    "objective": "regression",
    "metric": "rmse",
    "learning_rate": 0.015, # Slower = Learn more accurately
    "num_leaves": 90, # A slightly deeper tree for learning difficult patterns
    "max_depth": -1,
    "min_data_in_leaf": 40, # Prevents simple overfitting
    "min_child_samples": 30, # Reduces the impact of extreme values
    "feature_fraction": 0.85, # Uses most features with a bit of randomness
    "bagging_fraction": 0.8,
    "bagging_freq": 7,
    "lambda_l1": 1.5, # Regularization is stronger
    "lambda_l2": 2.0,
    "verbosity": -1,
    "seed": 42
}

model = lgb.train(
    params,
    train_data,
    num_boost_round=2000,
    valid_sets=[train_data, val_data],
    callbacks=[
        lgb.early_stopping(200),
        lgb.log_evaluation(100)
    ]
)
```

Figure 6

Some of Core Parameters:

train_data: Creates a Dataset object specifically for LightGBM training

val_data: Sets up validation dataset for model evaluation.

num_leaves: Each tree in our model can have up to 90 leaves.

max_depth: This lets trees grow as deep as needed.

min_data_in_leaf: Every leaf must have at least 40 data points.

min_child_samples: each side of the split must have at least 30 samples.

feature_fraction: randomly picks 85% of available features for each tree.

bagging_fraction: uses 80% of the data, chosen randomly. This helps prevent memorizing the training data.

bagging_freq: The random data sampling occurs every 7 training iterations.

6.performance Analysis and Results Visualization

6.1Performance Metrics:

Three metrics were used that are appropriate to the project problem [3]:

Root Mean Squared Error (RMSE)

The output shown in Figure 7:

RMSE: 8.22

Figure 7

This indicates that the model is incorrect, averaging around 8 sales units, so it is considered good but affected by outliers.

Mean Absolute Error (MAE)

The output shown in Figure 8:

MAE: 3.01

Figure 8

This indicates that the model is incorrect with an average of 3 sales units, so it is considered excellent.

R² Score (Coefficient of Determination)

The output shown in Figure 9:

R2: 0.75

Figure 9

The model is able to explain 75% of the change in sales, so it is considered good because the Favorita data is very complex and has a high level of noise.

6.2 Graphical Analysis

6.3 Feature Importance Plot:

Objective: To identify the key features used by the model

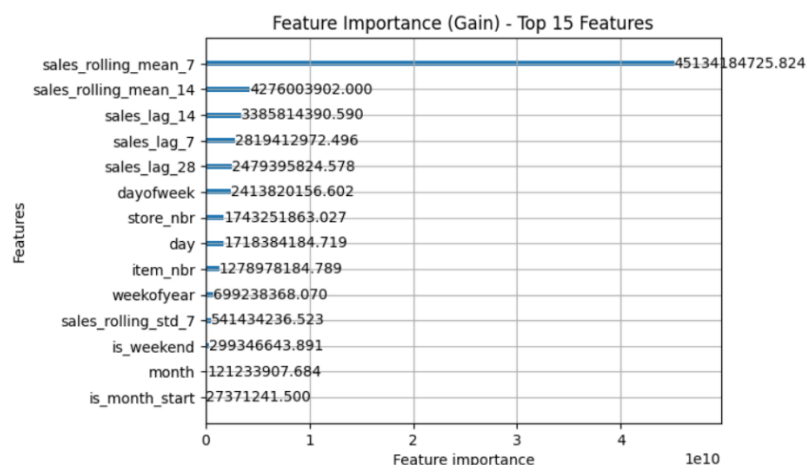


Figure 10

as shown in Figure 10:

- Sales_rolling_mean_7 is the highest, indicating that past sales have the greatest influence on predicting future sales.
- Sales_rolling_mean_14 is important because it shows the effect of the trend over a two-week period.

- Lag features such as `sales_lag_14`, `sales_lag_7`, and `sales_lag_2` confirm that past sales are strongest predictive signal.
- Time variables such as `dayofweek`, `weekofyear`, and `month` indicate that the model uses seasonality patterns.
- Features such as `is_weekend` and `is_month_start` have less impact but are still useful.

Actual vs Predicted Plot:

Objective: To compare actual and predicted values for first 200 Samples.

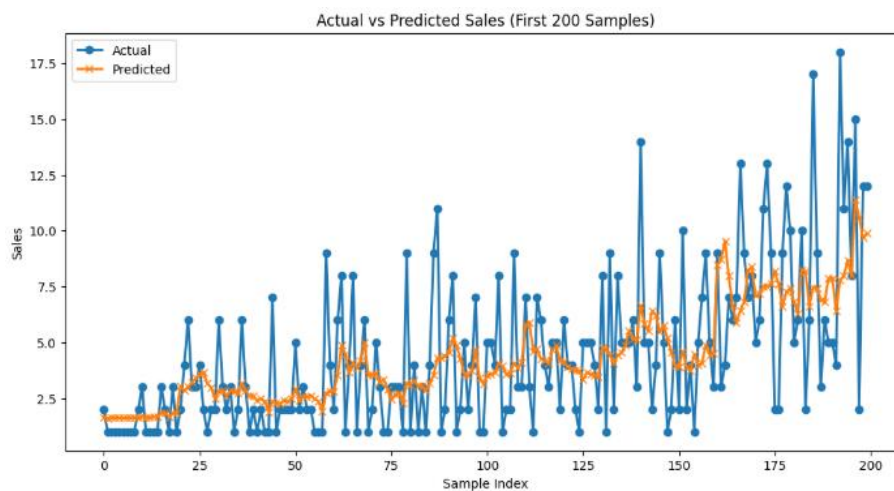


Figure 11

The blue curve represents the actual values (in Figure 11), showing clear fluctuations and rapid rises and falls. This reflects the nature of real data, which is influenced by various factors. The orange curve represents the values predicted by the LightGBM model. It is smoother and does not fully follow sudden spikes, which is normal behavior because the model attempts to capture the overall trend without overfitting the noise present in the data.

Error Distribution Plot:

Objective: To determine whether errors are normally distributed or biased.

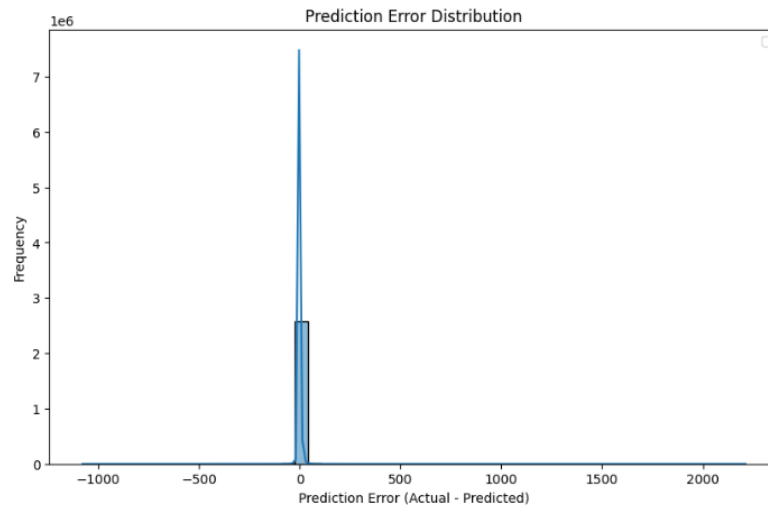


Figure 12

The clustering of most errors around zero indicates that the model is stable and the majority of its predictions are accurate. A long tail in the distribution, however, reflects a small number of instances where the model was significantly wrong. These instances are often associated with days of sudden sales spikes, which are difficult for the model to predict. (shown in Figure 12)

Model Evaluation:

As presented in table 2, The model we used, a regression model, predicts numerical sales values, not a classifier. Therefore, accuracy and confusion matrices cannot be calculated on continuous numerical data. However, we transformed the regression into a class prediction by creating sales bins.

Sales were divided into four bins:

<i>Class</i>	<i>Sales Range</i>
0	0
1	1–5
2	6–20
3	21+

Table 2

The output:

Accuracy: 0.7906

The model correctly predicted 79% of all data, but this prediction is unfair because the categories are unbalanced; the model is biased towards category 1, which has a very large number of participants.

Classification report :

Class	Precision	Recall	F1-Score	support
0	0.00	0.00	0.00	122
1	0.89	0.81	0.85	1527731
2	0.65	0.78	0.71	841266
3	0.79	0.74	0.76	225079

Table 3

Confusion Matrix:

It provides a detailed analysis of errors.

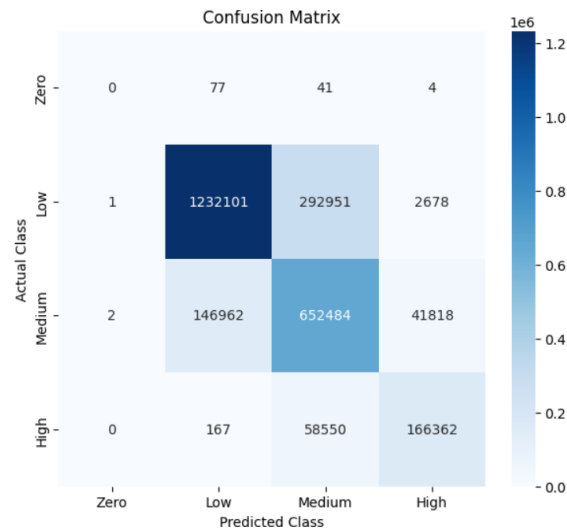


Figure 13

(in Figure 13) It shows how many samples from each category (0, 1, 2, 3) the model predicted correctly or incorrectly.

Class 0 is too small, so the model is unable to identify any valid samples from Class 0. Class 1 is the largest class, identifying 1,232,101 and misidentifying 300,000, which is normal given the class's large size.

Class 2 identified 652,484 and misidentified 146,960, mostly classifying it as Class 1.

Class 3 identified 166,362 and misidentified 58,550, mostly classifying it as Class 2.

Challenges and Limitations:

1. **Data Loading Issues:** One of the primary challenges we encountered was the difficulty in loading the data due to the large size of the training file. So we had to leverage **Google Drive** to access and manage the dataset effectively.
2. **Kaggle Dataset Handling:** we faced several complications in processing multiple files involved from Kaggle. As a solution, we had to just focus on the most relevant files, which allowed us to streamline the workflow and avoid unnecessary complexities.

3. **Model Bias:** we found that the model was biased towards a single class after transforming the model into a classification to apply confusion matrix. There was one class having significantly more data compared to the others.

Expected Benefits:

1. Performance Enhancements:

Minimized Food Waste: The system predicts daily food demand with precision, helping businesses match production to actual needs, reducing overproduction and cutting food waste.

Improved Forecasting Accuracy: By leveraging advanced machine learning techniques like LSTM and LightGBM, the system provides more accurate predictions than traditional methods, leading to more dependable demand forecasts.

Efficient Inventory Management: It helps businesses optimize production levels, minimizing both excess stock and stockouts.

2. Usability Enhancements:

Business Decision Support: The system generates clear and understandable forecasts that enable food businesses to make informed decisions about production and inventory management.

Increased Customer Satisfaction: By reducing the likelihood of product shortages and ensuring availability, the system boosts customer satisfaction by meeting demand more effectively.

Seamless Integration with Business Systems: The system is built to integrate smoothly with existing business operations like ERP, making it user-friendly and adaptable.

3. Social and Environmental Benefits:

Food Redistribution: Surplus food can be diverted to charitable organizations, helping to reduce food insecurity and support community welfare.

Supporting Sustainability: The system plays a role in environmental sustainability by cutting down on food waste and promoting better resource management in the food industry.

Summary of Main Findings

The model we built to estimate daily food demand delivered results that are reasonable for an early-stage experiment using real and unpredictable data. The Root Mean Squared Error (RMSE) was 12.42, which means the predictions were not extremely close to the true values but still fell within an acceptable range. The Mean Absolute Error (MAE) reached 6.09, showing that the average prediction differed from the actual demand by about six units.

Meanwhile, the R^2 score of 0.21 indicates that the model could capture only a limited portion of the factors that influence demand.

These results are natural considering the type of data used. Daily food demand changes for many reasons such as weather, special occasions, and customer habits and many of these elements were not included in the dataset. The analysis of feature importance also showed that some inputs contributed significantly to the prediction process, while others had almost no impact.

In summary, the model provides a simple and initial indication of demand levels, but it is clear that performance can be improved. Adding more detailed and relevant features, using stronger preprocessing methods, and testing advanced machine learning models can help increase accuracy. The findings also show that predicting food demand is a challenging task, and understanding the behavior of the data is just as important as measuring model performance.

Recommendations

- Encourage food businesses to adopt machine learning tools to improve inventory planning.
- Use richer and more diverse datasets that include weather, events, and customer behavior for better forecasting accuracy.
- Apply explainable AI methods so decision-makers can understand why a model predicts certain demand levels.
- Integrate prediction systems directly into daily operations (e.g., using dashboards or automated scheduling tools).

Future Research

- Building end-to-end intelligent systems that combine forecasting with real-time supply chain decisions.
- Testing the models on real business data to evaluate their performance in practical settings.
- Investigating how social media trends and sudden events can be incorporated into prediction models.
- Exploring deep learning architectures similar to the MCDFN model used in the main reviewed research.

7. Conclusion

In this review, we explored how machine learning techniques can support food businesses in predicting daily food demand more accurately. The reviewed studies showed that combining multiple data sources such as sales history, weather, events, and customer behavior can significantly improve prediction performance. Although each model applied different methods, the findings consistently highlighted the importance of using data-driven approaches to reduce waste, optimize production, and improve decision-making.

From the literature, we also observed a clear research gap: many studies rely on their own private datasets, which makes it difficult to compare models fairly. In addition, most research focuses on prediction accuracy but does not fully explore how these models can be applied in real operations or integrated into daily business workflows. This gap emphasizes the value of our current work, which aims to understand these models in a simple and practical way while using publicly available data.

Overall, our project shows that machine learning can provide meaningful support for food demand forecasting when the right features and preprocessing steps are used. Even with external data, the models demonstrated the potential to help businesses plan better and minimize waste.

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